DEVSECOPS LAB BOOK · PHASE 5: ADVANCED TOPICS

Chapter 21

Disaster Recovery

Backup, Restore & Recovery Planning

Complete backup strategy for the entire platform.
Backup everything. Test recovery procedures. Meet RPO/RTO requirements.

TIME ESTIMATE: 4–5 HOURS

21.0 · CHAPTER GOAL

What You'll Walk Away With

Instructor Framing

Teach this chapter as the platform's insurance policy. Every prior chapter builds capability — this one guarantees it survives a disaster. Emphasize that RPO/RTO targets drive every design decision that follows.

KEY TERMS TO ANCHOR

RPO — Recovery Point Objective — max acceptable data loss

RTO — Recovery Time Objective — max acceptable downtime

- Complete backup strategy for all services
- Tested recovery procedures
- RPO/RTO metrics defined
- Disaster recovery runbook
- Incident response integration

Cold, Warm & Hot DR

Tier 1

Cold DR

Backup only. Manual restore, rebuild infra, restore data, repoint DNS.

RECOVERY TIME

4–8 HRS

UNACCEPTABLE

Tier 2

Warm DR

Replication + automatic failover, detected by HA system. Backup as safeguard.

RECOVERY TIME

1–5 MIN

ACCEPTABLE

Tier 3

Hot DR

Full multi-region duplication with automatic failover on disaster.

RECOVERY TIME

< 1 SEC

ZERO DOWNTIME

What to Backup, by Priority

Tier 1

Critical

RPO: 1 hour

- Vault data & encryption keys
- Database (PostgreSQL, MySQL)
- Infrastructure code (Git repos)
- Configuration files (/etc)

Tier 2

Important

RPO: 24 hours

- Container registry images
- DNS zone files
- LDAP directory
- Certificates & keys

Tier 3

Optional

RPO: 7 days

- Logs & audit trails
- CI/CD history
- Old container images

Cadence, Coverage & Retention

HOURLY	**Tier 1 — Critical**	Vault data snapshot · Database transaction logs	*Keep 24 copies (rolling)*
DAILY	**Tier 1 & 2 — Full backup**	Full database backup · configs · container images	*Keep 30 copies (monthly rotation)*
WEEKLY	**Archive**	Full platform snapshot to long-term storage	*Keep 52 copies (yearly)*
MONTHLY	**Compliance**	Full audit trail export · compliance evidence package	*Immutable archive (3+ years)*

Hourly Snapshot → Encrypt → Offsite



RESTORE PROCEDURE (Vault Data Loss)

1. Stop Vault service
2. Clear corrupted storage directory
3. Decrypt & restore snapshot via `vault operator raft snapshot restore`
4. Unseal Vault
5. Verify with `vault status`

PostgreSQL: Base Backup + PITR

Backup Strategy

- Daily full backup via pg_basebackup (tar + gzip)
- Continuous WAL archiving enabled for point-in-time recovery
- archive_mode = on in postgresql.conf
- Synced to S3 offsite storage
- Retention: 30 daily backups kept locally

Point-in-Time Recovery

1. Stop PostgreSQL service
2. Remove current (corrupted) data directory
3. Restore full backup from latest tarball
4. Set recovery_target_time in recovery.conf
5. Start PostgreSQL — WAL replays to target time

Result: data consistent as of the chosen recovery_target_time.

Git Repositories Are Your Backup

What Lives in Git

- Packer HCL templates
- OpenTofu code
- Ansible playbooks
- Kubernetes manifests
- CI/CD configurations
- Documentation

Rebuild the entire infrastructure from Git history alone.

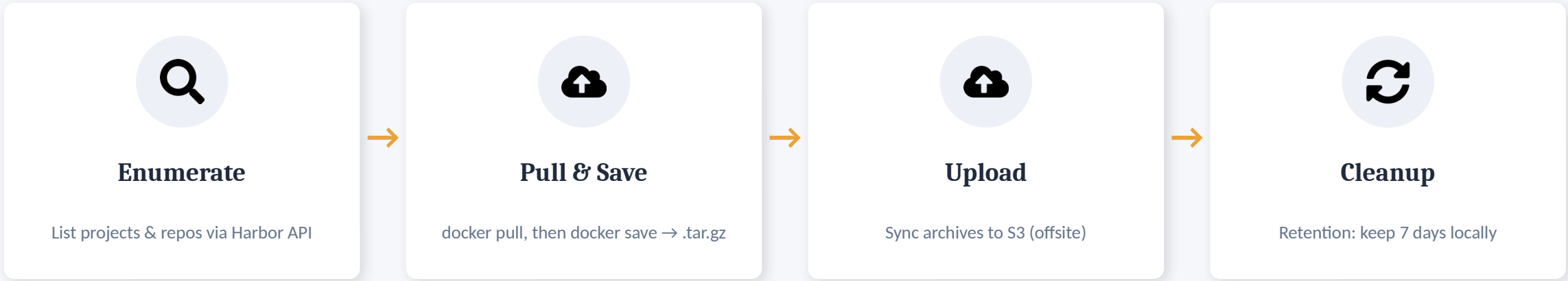
Offsite Mirror Strategy




```
git push backup1/backup2/backup3 --all
```

Push to multiple remotes so no single provider outage causes loss.

Harbor Export, Save & Restore



RESTORE FROM BACKUP

```
for backup_file in /mnt/backups/harbor/*.tar.gz; do
  docker load -i $backup_file
  docker push $(tar -tzf $backup_file | head -1)
done
```

RTO for Harbor: 10–30 minutes · RPO: < 1 hour

Primary Datacenter Down — Response Timeline

PHASE 1

Assessment

15 MIN

- Confirm disaster via health check
- Page on-call + backup on-call
- Open incident, spin up war room

PHASE 2

Failover

30 MIN

- Promote DR infra, repoint DNS
- Start DR services (Ansible)
- Restore DB from latest backup

PHASE 3

Validation

15 MIN

- Run health checks (Ch.13–20)
- Verify data & API health
- Confirm Git repos reachable

PHASE 4

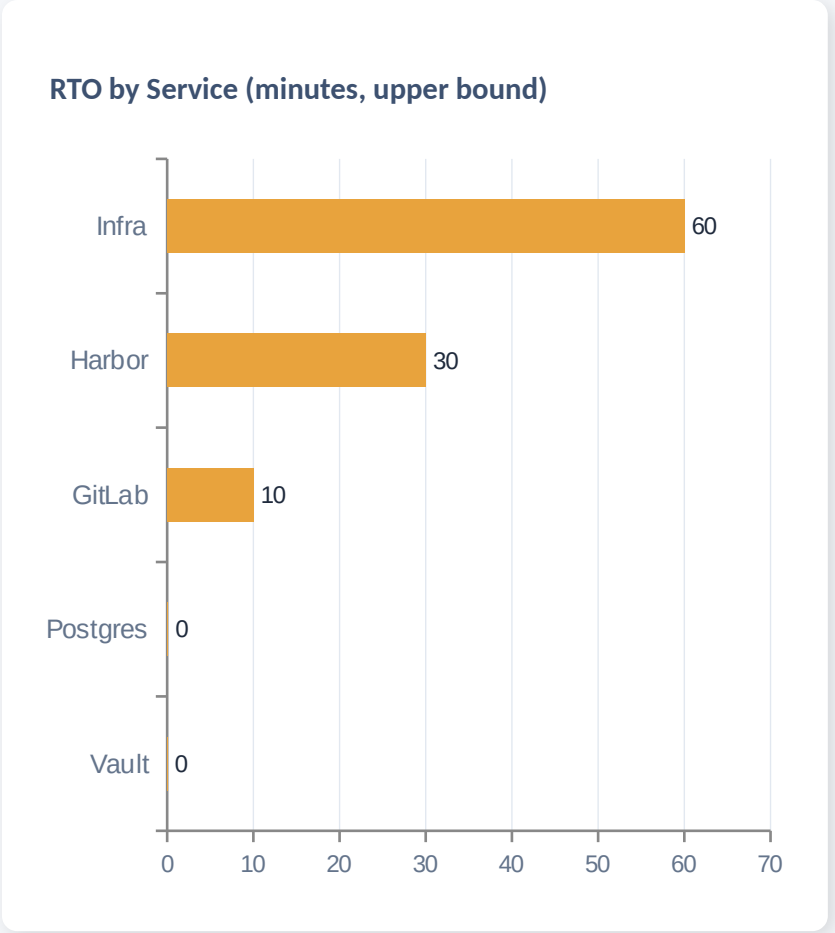
Communication

ONGOING

- Status update every 15 min
- #incident-response channel
- Root-cause investigation begins

After Implementing HA + DR

Service	RPO	RTO	Recovery Mechanism
Vault	0s	**<1s**	HA clustering
PostgreSQL	0s	**1–5s**	Replication + Patroni
GitLab	<5m	**5–10m**	Backup + restore
Harbor	<1h	**10–30m**	Image export
Infrastructure	<5m	**30–60m**	Git + IaC rebuild



Total end-to-end RTO across the full DR runbook: 60 minutes.

Before & After DR Planning

Phase	Time	Tasks
Assessment	15 min	Confirm disaster, contact team
Failover	30 min	Promote DR, restore data
Validation	15 min	Health checks, verify data
Communication	Ongoing	Update stakeholders

BEFORE DR PLANNING

- No backups (data loss certain)
- No documented procedures (chaos)
- Unknown recovery time (hours/days)

AFTER DR PLANNING

- Automated backups (hourly)
- Documented procedures (runbooks)
- Known recovery time (< 1 hour)
- Tested recovery procedures (confidence)



KEY TAKEAWAY

Preparation Ensures Recovery

Total RTO

60 MINUTES

from disaster confirmation to full service recovery